A translated Laguerre criterion in the positive-Gaussian case

Matthew J. Colbrook*

Department of Applied Mathematics and Theoretical Physics
University of Cambridge, Cambridge CB3 0WA, United Kingdom
m.colbrook@damtp.cam.ac.uk

7 November 2026

Abstract

The strict translated Laguerre criterion is proved when the Gaussian coefficient is positive. The case of Gaussian coefficient zero remains unresolved here.

Catalogue scope. AIM 694; partial progress. Canonical target at revision `1e56463`.

1 Statement and argument

Status: partial only; this problem is not counted as resolved. We prove the stated implication when the coefficient a of the explicit Gaussian factor in the canonical product is strictly positive. The case $a = 0$ remains unresolved here.

For a real entire function f, write

$$U(x,y) = \operatorname{Re} f(x+iy), \qquad V(x,y) = \operatorname{Im} f(x+iy).$$

Since f is real on the real axis, the translated function in the question is $f_y(x) = 2U(x,y)$. Its strict Laguerre inequality is therefore equivalent to

$$U_x(x,y)^2 - U(x,y)U_{xx}(x,y) > 0 \qquad (x \in \mathbb{R}, y > 0). \tag{1}$$

Lemma 1.1 (A horizontal-decay criterion)**.** *Suppose that a real entire function f satisfies (1) and, for every finite $Y > 0$,*

$$\sup_{|y| \le Y} |f(x+iy)| \longrightarrow 0 \qquad \text{as } x \longrightarrow \pm\infty. \tag{2}$$

Then every zero of f is real.

Proof. Suppose to the contrary that $f(x_0 + iy_0) = 0$ with $y_0 > 0$. At any point of the nodal set $U = 0$ in the upper half-plane, (1) implies $U_x \neq 0$. The implicit-function theorem therefore produces a local graph $x = x(y)$ through (x_0, y_0) on which $U = 0$. Continue this graph downward over its maximal interval $(\beta, y_0]$, where $0 \le \beta < y_0$. Equivalently, it is the maximal solution of

$$x'(y) = -\frac{U_y(x(y),y)}{U_x(x(y),y)}, \qquad x(y_0) = x_0,$$

restricted to the nodal set. Local uniqueness permits this continuation as long as the graph remains in a bounded part of the upper half-plane.

*The solutions and partial results in this document were found by AI. The supplied research and the separate repository review are AI work; human peer review and formal verification are not claimed.

Along the graph U_x is continuous and never zero, hence has constant sign. The Cauchy–Riemann equations give

$$\frac{d}{dy}V(x(y),y) = \frac{U_y(x(y),y)^2 + U_x(x(y),y)^2}{U_x(x(y),y)}. \tag{3}$$

Thus $V(x(y),y)$ is strictly monotone, with derivative of one fixed sign, and its value at y_0 is zero. In particular, after fixing any $y_1 \in (\beta, y_0)$, all values at $\beta < y \le y_1$ have the same sign and have absolute value at least $|V(x(y_1),y_1)| > 0$.

If $\beta > 0$ and there were a bounded sequence $x(y_j)$ with $y_j \downarrow \beta$, a subsequence would converge to x_*. Continuity would give $U(x_*,\beta) = 0$ and (1) would give $U_x(x_*,\beta) \neq 0$. The implicit-function theorem at this point gives a unique graph on a whole neighborhood of β. The original graph, which meets that neighborhood at $(x(y_j),y_j)$, agrees with it by local uniqueness and extends below β. This contradicts maximality. Consequently $|x(y)| \to \infty$ as $y \downarrow \beta$ when $\beta > 0$. Condition (2), with $Y = y_0$, then gives $V(x(y),y) \to 0$, contradicting the lower bound just established.

It remains to consider $\beta = 0$. Take a sequence $y_j \downarrow 0$. If $x(y_j)$ has a bounded subsequence, pass to a further convergent subsequence. Since f is real on the real axis, continuity gives $V(x(y_j),y_j) \to 0$ on that subsequence. If no bounded subsequence exists, pass to a subsequence with $|x(y_j)| \to \infty$ and use (2) to reach the same conclusion. Either alternative contradicts (3). There are therefore no zeros in the upper half-plane. Conjugation excludes zeros in the lower half-plane. □

Proposition 1.2. *Let f satisfy the hypotheses of Problem 694 and have the canonical product*

$$f(z) = Ce^{-az^2+bz}z^m \prod_k \left(1 - \frac{z}{z_k}\right) e^{z/z_k}, \qquad \sum_k |z_k|^{-2} < \infty,$$

as in [1, Definition 1.1]. If $a > 0$, then every zero of f is real and f belongs to the Laguerre–Pólya class.

Proof. The strict inequality excludes $f \equiv 0$. Set $E_1(w) = (1-w)e^w$. There is an absolute constant C_0 such that

$$\log|E_1(w)| \le C_0|w|^2 \qquad (w \in \mathbb{C}).$$

For $|w| \le 1/2$ this follows by expanding $\log(1-w)+w$; for $|w| > 1/2$ use $\log|1-w| + \operatorname{Re} w \le \log(1+|w|) + |w|$. Fix $\varepsilon > 0$ and split the canonical product into a finite head and a tail with $C_0 \sum_{k>K} |z_k|^{-2} < \varepsilon$. The logarithm of the modulus of the finite head, including $Ce^{bz}z^m$, is at most $O(|z| + \log(2+|z|))$, uniformly in the argument of z. The tail is at most $\varepsilon|z|^2$ in logarithmic modulus. It follows that

$$\log\left|Ce^{bz}z^m \prod_k E_1(z/z_k)\right| \le o(|z|^2)$$

as an upper bound uniform in the argument. More precisely, for every $\varepsilon > 0$ this logarithm is at most $C_\varepsilon + \varepsilon|z|^2$.

For $z = x + iy$ with $|y| \le Y$, choose $0 < \varepsilon < a$. Then

$$\log|f(x+iy)| \le C_\varepsilon - (a-\varepsilon)x^2 + (a+\varepsilon)Y^2,$$

which proves (2). The preceding lemma excludes all nonreal zeros. The canonical-product characterization in [1, Definition 1.2] now gives $f \in \mathcal{LP}$. □

The proof uses horizontal decay supplied by $a > 0$. It does not provide that decay when $a = 0$, and it does not resolve the full conjecture.

References

[1] G. Csordas and A. Escassut, *The Laguerre inequality and the distribution of zeros of entire functions*, Ann. Math. Blaise Pascal **12** (2005), 331–345. https://doi.org/10.5802/ambp.210.